



Scilab program & Implementation (20)	Waveforms and tabulation (20)	calculation and result (20)	viava- voce (20)	Result/ observat ⁿ (20)
20	20	20	15	20
				95

1/8/26

exp 4

FREQUENCY

MODULATION:-

AIM:-

To stimulate frequency modulation in Scilab and to calculate beta.

EQUIPMENT REQUIRED:-

Scilab experiment

CODE:-

```
clc;
```

```
clear;
```

```
close;
```

```
Am = 2;
```

```
Ac = 3;
```

```
fm = 5;
```

```
fc = 50;
```

```
beta = 5;
```

```
t = 0:0.0001:1;
```

```
Em = Am * cos(2 * %pi * fm * t);
```

```
Ec = Ac * cos(2 * %pi * fc * t);
```

```
EFM = Ac * cos(2 * %pi * fc * t + beta * sin(2 * %pi * fm * t));
```

```
subplot(3,1,1);
```

```
plot(t, Em);
```

```
title("Message Wave");
```

S.No	Particulars	Theory		Practical		
		Amplitude	Frequency	Amplitude	Time	Frequency
1.	Message signal	2	5	2	$T = x_2 - x_1$ $\Rightarrow 0.30 - 0.10$ $= 0.20s$	$F = \frac{1}{T}$ $\Rightarrow \frac{1}{0.20} \Rightarrow 5Hz$
2.	Carrier signal	3	50	3	$T = x_2 - x_1$ $\Rightarrow 0.04 - 0.02$ $= 0.02s$	$F = \frac{1}{T}$ $\Rightarrow \frac{1}{0.02} \Rightarrow 50Hz$
3.	FM signal	—	—	3	$t_{min} \Rightarrow$ $0.04 - 0.02 \Rightarrow 0.02s$ $T_{max} \Rightarrow$ $0.3 - 0.32$ $\Rightarrow 0.04s$	$\rightarrow F = \frac{1}{T_{min}}$ $= \frac{1}{0.02} \Rightarrow 50Hz$ $F = \frac{1}{T_{max}} \Rightarrow 25Hz$ $\Rightarrow 40s$
					T_{max} T_{min}	$f_{min} = \frac{1}{T_{max}}$ $f_{max} = \frac{1}{T_{min}}$

subplot (3,1,2);
plot (t, Ec);
title ("carrier Wave");

subplot (3,1,3);
plot (t, EFM);
title ("Frequency modulated Wave");

CALCULATION:-

$$\Delta F = F_c - F_{\min}$$
$$= 50 - 25$$
$$\Delta F = 25 \text{ Hz}$$

$$f_m = 5 \text{ Hz}$$

$$f_c = 50 \text{ Hz}$$

$$\beta = 5$$

$$\beta = \frac{\Delta f}{f_m} \Rightarrow \Delta f = \beta f_m$$
$$= 5 \times 5$$
$$= 25 \text{ Hz}$$

$$\Delta F = F_{\max} - F_c$$
$$= 75 - 50$$
$$= 25 \text{ Hz}$$

$$f_{\min} = f_c - \Delta f$$
$$= 50 - 25$$
$$= 25 \text{ Hz}$$

$$\rightarrow \beta = \frac{\Delta F}{f_m} \Rightarrow \frac{25}{5} \Rightarrow 5 \text{ Hz}$$

Verification

$$\beta = \frac{f_{\max} - f_c}{f_m}$$

$$\beta = 5$$

$$f_{\max} = f_c + \Delta f$$
$$= 50 + 25$$
$$= 75 \text{ Hz}$$

RESULT:-

Frequency modulation is generated and simulated
thus, beta is calculation.

~~Q3~~